

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

TOPIC: CLOUD COMPUTING AND CLOUD TECHNOLOGIES

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Introduction to Cloud Computing

Cloud Computing is a technology that provides computing resources and services over the Internet. These resources include computing power, storage, databases, networking, software, analytics and artificial intelligence services.

In traditional computing, organizations need to purchase physical servers, storage devices and networking equipment. They are also responsible for maintenance, upgrades and security. Cloud computing allows organizations to use these resources from a cloud provider without managing all the physical infrastructure themselves.

For example, when we use Google Drive, our files can be stored on remote cloud infrastructure and accessed from different devices through the Internet.

Cloud computing is widely used in websites, mobile applications, e-commerce, education, banking, entertainment, healthcare and software development.

Definition

Cloud Computing is the delivery of computing resources and services over the Internet on demand.

Key Characteristics

- **On-Demand Self-Service** - Users can obtain computing resources whenever required.
- **Broad Network Access** - Cloud services can be accessed through computers, smartphones, tablets and other network-connected devices.
- **Resource Pooling** - Cloud providers share their physical resources among multiple customers using technologies such as virtualization.
- **Rapid Elasticity** - Resources can be increased or decreased according to demand.
- **Measured Service** - Resource usage can be monitored and measured, allowing usage-based management and billing.

Importance

- Reduces infrastructure requirements.
- Provides easy access to data and services.
- Supports remote work and collaboration.
- Provides scalability.
- Makes application deployment faster.
- Supports backup and recovery.

Cloud Technology and How It Works

Cloud Technology refers to the collection of technologies, platforms and services used to build and operate cloud-based systems.

Cloud computing is the concept or model, while cloud technologies are the actual technologies and services used to implement it.

Examples of cloud technologies include:

- Virtual Machines
- Cloud Storage
- Cloud Databases
- Containers
- Serverless Computing
- APIs
- Virtual Networks
- Cloud Security
- Artificial Intelligence services

Examples of cloud technology platforms are:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud

Cloud Computing vs Cloud Technology

Cloud Computing	Cloud Technology
1. Concept of delivering computing services over the Internet.	1. Technologies used to provide cloud services.
2. Explains how cloud services are provided and consumed.	2. Explains the tools and platforms used to provide services.
3. Includes service models like IaaS, PaaS, and SaaS.	3. Includes virtualization, containers, APIs, and automation.
4. Includes deployment models like Public, Private, and Hybrid Cloud.	4. Includes platforms such as AWS, Azure, and Google Cloud.
5. Provides computing, storage, networking, and database services.	5. Uses servers, data centers, networks, and storage systems.

6. Allows users to access resources on-demand.	6. Enables scalability, resource sharing, and efficient management.
7. Follows a pay-as-you-use or subscription model.	7. Supports security, monitoring, and automation of cloud resources.
8. Example: Using Google Drive or Gmail through the Internet.	8. Example: Virtual machines, Docker, Kubernetes, AWS, and Azure.

How Cloud Computing Works

Cloud computing allows users to access computing services such as **servers, storage, databases, applications, and processing power through the Internet**. Users can access these services without owning or maintaining the physical infrastructure.

The basic process is:

User Device → Internet → Cloud Platform → Cloud Resources → Result

1. **User sends a request:** The user opens a cloud application or service on a computer, smartphone, or other device.
2. **Request travels through the Internet:** The request is securely transmitted to the cloud provider.
3. **Cloud platform processes the request:** The cloud platform identifies the required service and resources.
4. **Resources are allocated:** Required computing power, memory, storage, or database resources are assigned.
5. **Request is processed:** Cloud servers perform the required operation and generate the result.
6. **Result is returned:** The processed information is sent back to the user's device through the Internet.

Cloud providers operate **large data centers** containing servers, storage systems, and networking equipment. These resources are managed using specialized software, **virtualization, containers, and automation**. Virtualization allows multiple users or applications to share physical resources efficiently.

Example: When a user uploads a file to Google Drive, the file travels from the user's device through the Internet to the cloud platform, where it is stored on cloud servers. The user can later access the file from any device with an Internet connection.

Cloud Architecture and Virtualization

Cloud architecture describes the components that work together to deliver cloud services.

Main Components

- **Front End** - The user-facing side, such as a web browser, mobile application, desktop application or user interface.
- **Back End** - The infrastructure maintained by the cloud provider, including servers, storage, databases, virtual machines, networking, security systems and management software.
- **Internet / Network** - Connects users to cloud infrastructure.

Basic Architecture

USER	INTERNET	CLOUD PLATFORM
Browser	Requests	Compute
Mobile App	Authentication	Storage
Desktop App	Data Transfer	Database
		Networking
		Security

Virtualization

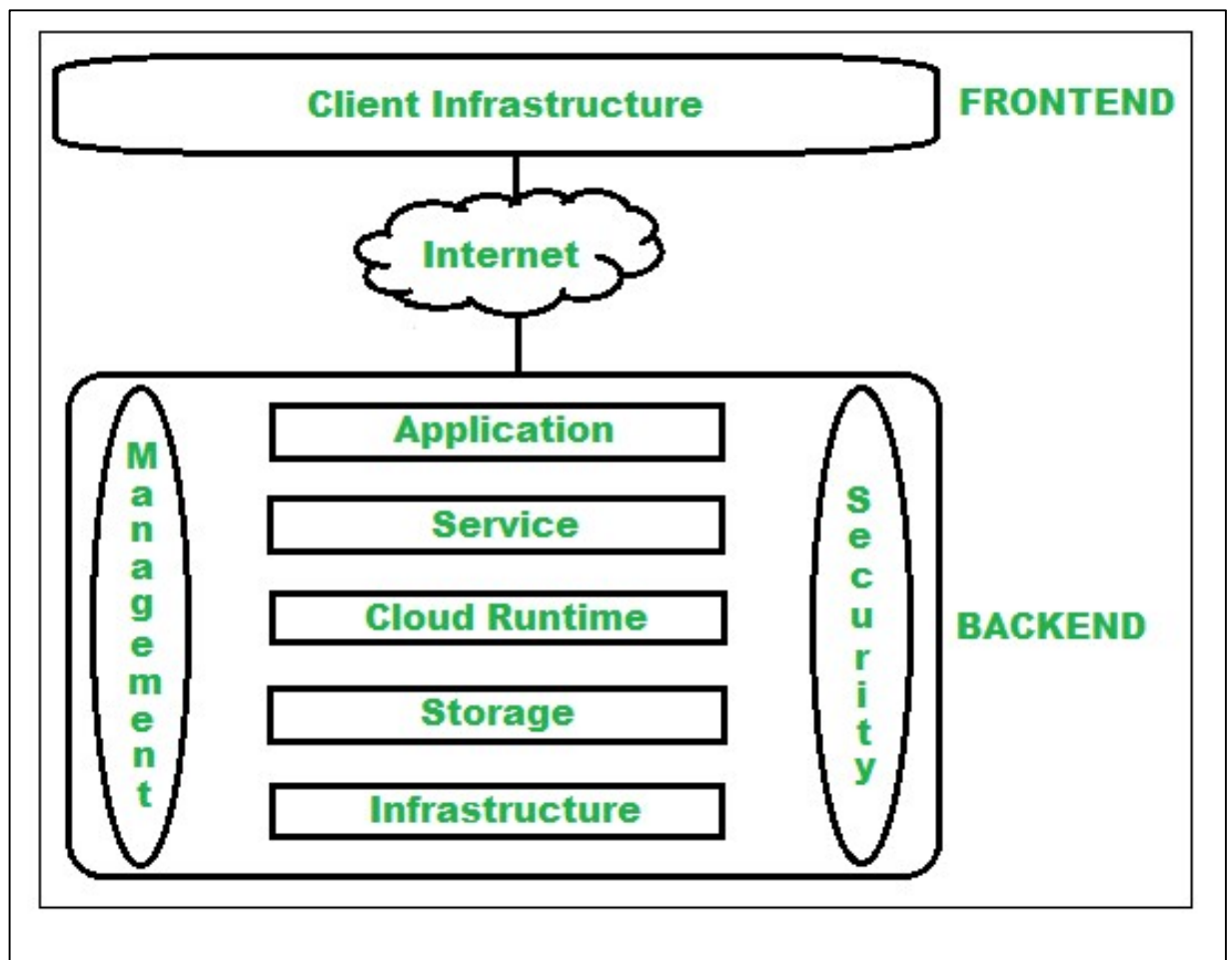
Virtualization is one of the important technologies behind cloud computing. It allows a single physical server to run multiple Virtual Machines (VMs).

Physical Server → Virtual Machine 1 | Virtual Machine 2 | Virtual Machine 3

Each virtual machine can run its own operating system and applications.

Advantages of Virtualization

- Better hardware utilization
- Reduced infrastructure cost
- Easy resource allocation
- Isolation between workloads
- Faster deployment
- Supports cloud scalability



Cloud Service and Deployment Models

Cloud Service Models

There are three major cloud service models.

4.1 IaaS — Infrastructure as a Service

IaaS provides basic computing infrastructure such as virtual machines, storage, networking and processing resources. The user manages the operating system, applications and data.

Example: AWS EC2, Azure Virtual Machines.

4.2 PaaS — Platform as a Service

PaaS provides a development platform for building and deploying applications. The provider manages the infrastructure and runtime environment. Developers mainly focus on application code and data.

Example: Azure App Service and similar managed application platforms.

4.3 SaaS — Software as a Service

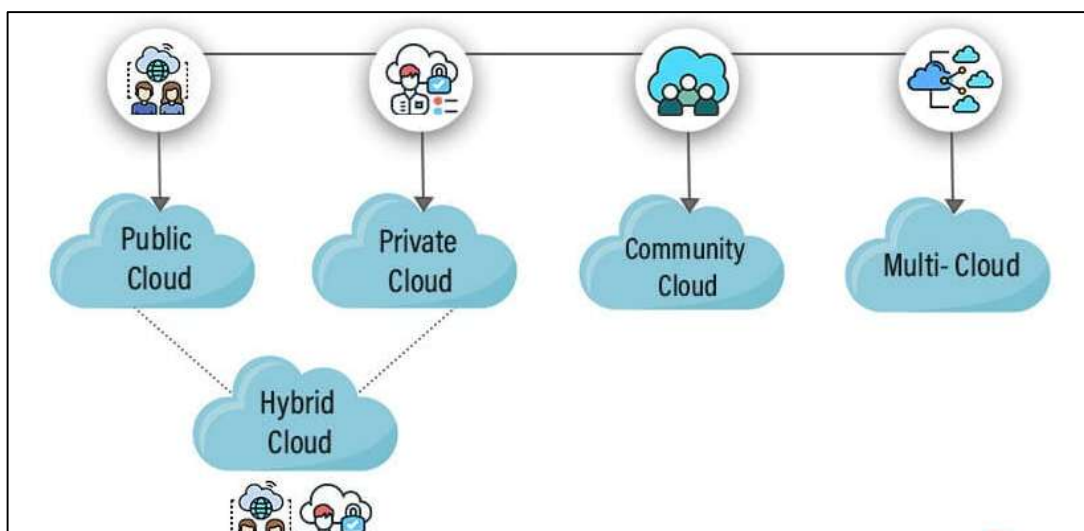
SaaS provides complete software applications through the Internet. Users do not need to install or maintain the underlying infrastructure.

Examples: Gmail, Google Docs, Microsoft 365.

Easy Memory Trick: IaaS → Infrastructure | PaaS → Platform | SaaS → Software

Cloud Deployment Models

- **Public Cloud** — Cloud infrastructure is operated by a provider and shared among multiple customers.
- **Private Cloud** — Cloud infrastructure is dedicated to a single organization.
- **Hybrid Cloud** — Combination of public and private cloud environments.
- **Community Cloud** — Infrastructure shared by organizations with common requirements.
- **Multi-Cloud** — Infrastructure shared by organizations with common requirements.



Amazon Web Services (AWS)

Amazon Web Services (AWS) is a major cloud computing platform provided by Amazon. AWS provides a large collection of cloud services that organizations can use for computing, storage, databases, networking, security, analytics and artificial intelligence.

Instead of purchasing physical servers, organizations can use AWS resources according to their requirements.

Important AWS Services

1. Amazon EC2

EC2 stands for Elastic Compute Cloud. It provides virtual servers that can run applications and websites. Use: Website hosting and application deployment.

2. Amazon S3

S3 stands for Simple Storage Service. It is used to store files and objects such as images, videos, documents and backups.

3. Amazon RDS

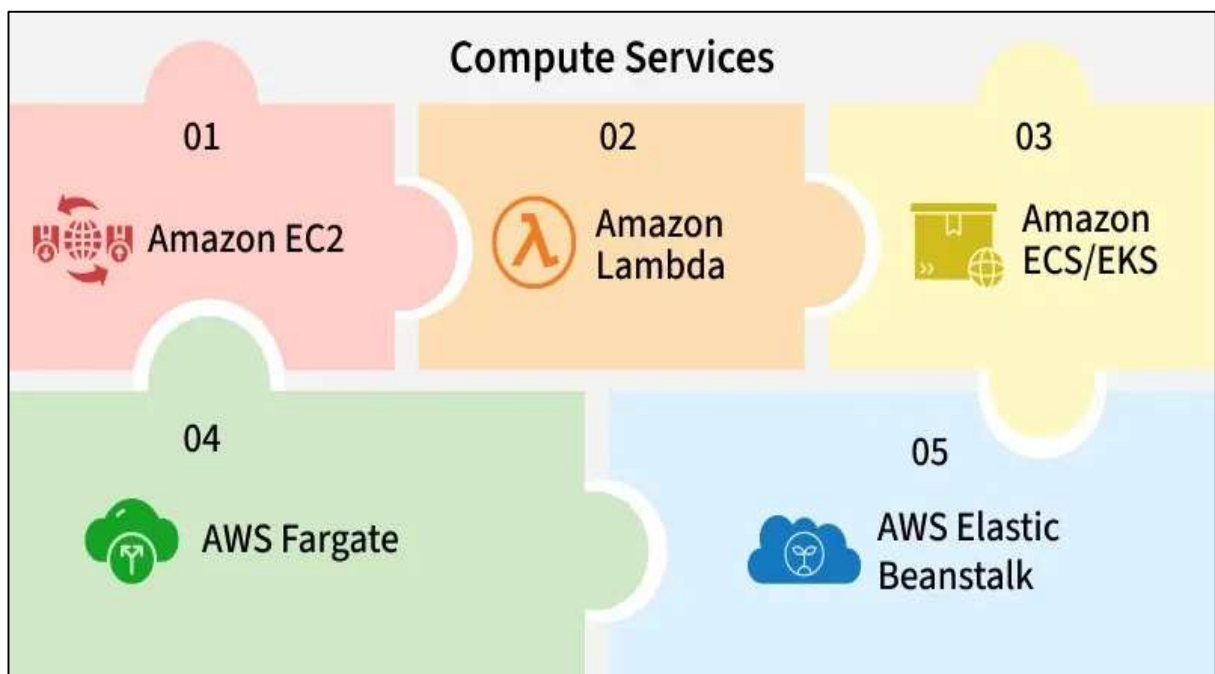
RDS stands for Relational Database Service. It provides managed relational databases.

4. AWS Lambda

AWS Lambda is a serverless computing service that allows developers to execute code without directly managing servers.

5. Amazon VPC

VPC stands for Virtual Private Cloud. It provides an isolated virtual networking environment for AWS resources.



AWS Applications

- Website hosting
- Mobile applications
- E-commerce
- Data storage
- Backup
- Software development
- AI/ML applications

Advantages of AWS

- Large number of services
- Scalable infrastructure
- Flexible resource usage
- Global infrastructure

Microsoft Azure

Microsoft Azure is Microsoft's cloud computing platform. It provides cloud services for computing, storage, databases, networking, security, analytics, artificial intelligence and application development.

Azure allows organizations to build, deploy and manage applications using Microsoft's cloud infrastructure.

Important Azure Services

1. Azure Virtual Machines

Azure Virtual Machines provide virtual computers in the cloud. They can be used to run websites, applications, databases and development environments.

2. Azure Blob Storage

Blob Storage is used for storing large amounts of unstructured data such as images, videos, documents and backup files.

3. Azure SQL Database

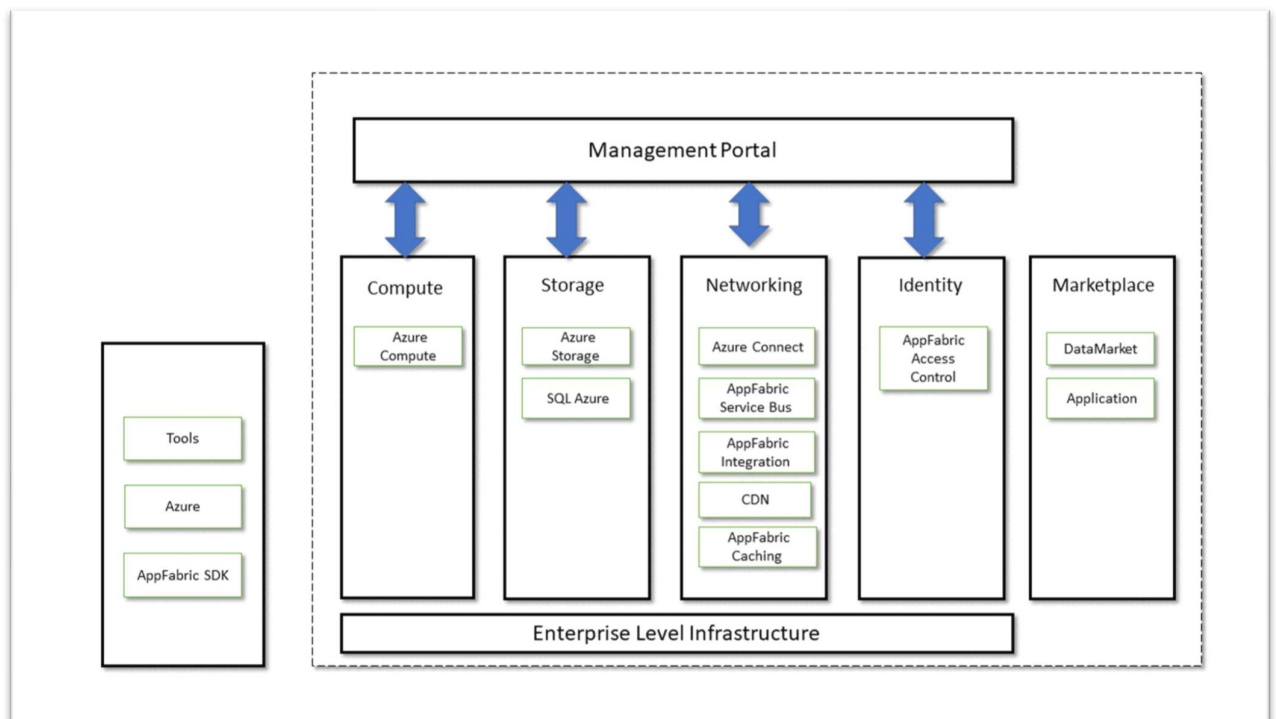
Azure SQL Database is a managed relational database service based on Microsoft SQL technologies.

4. Azure Functions

Azure Functions is a serverless computing service used to execute code in response to events.

5. Azure Virtual Network

It provides networking capabilities for communication between Azure resources.



Applications of Azure

- Web applications
- Enterprise applications
- Data analytics
- Artificial Intelligence
- Software development
- Backup and disaster recovery

Advantages of Azure

- Strong integration with Microsoft technologies
- Wide range of cloud services
- Scalable infrastructure
- Security and identity services
- Developer-friendly tools

Google Cloud and Other Cloud Technologies

Google Cloud is Google's cloud computing platform. It provides services for computing, storage, databases, networking, data analytics, Artificial Intelligence and Machine Learning.



Important Google Cloud Services

- Compute Engine: Virtual machines for running applications.
- Cloud Storage: Object storage for files and data.
- Cloud SQL: Managed relational database service.
- Google Kubernetes Engine (GKE): Managed Kubernetes service for containerized applications.

Uses

- Data analytics
- AI and Machine Learning
- Web applications
- Cloud-native development
- Big Data processing

Other Important Cloud Technologies

1. Containers

Containers package an application together with its required dependencies. They make applications easier to develop, test and deploy consistently. Example: Docker.

2. Kubernetes

Kubernetes is a platform used to manage containerized applications. It helps with deployment, scaling, load management and container orchestration.

3. Serverless Computing

Serverless computing allows developers to run application code without directly managing servers. Examples: AWS Lambda, Azure Functions.

4. Cloud Storage

Cloud storage allows users to store files on remote infrastructure and access them through a network.

5. Cloud Databases

Cloud databases provide database functionality without requiring organizations to manage all database infrastructure themselves.

Applications of Cloud Technology

Cloud computing and cloud technologies are used in many industries.

1. Education

Cloud technology supports online classes, digital libraries, online examinations, file sharing and learning platforms.

2. E-Commerce

E-commerce websites use cloud infrastructure for website hosting, product databases, product images, customer data, order processing and payment systems. Cloud scalability helps handle increased traffic during sales.

3. Banking and Finance

Cloud technologies can support data processing, online banking applications, analytics, backup and digital services. Sensitive financial systems require strong security and compliance controls.

4. Healthcare

Cloud services can support data storage, digital applications, collaboration and data analysis. Healthcare organizations must follow applicable privacy and regulatory requirements.

5. Entertainment

Streaming platforms use cloud infrastructure to store and deliver movies, music, videos and games.

6. Software Development

Developers use cloud platforms for development, testing, databases, hosting, CI/CD and application deployment.

7. Artificial Intelligence

Cloud platforms provide scalable computing resources for Machine Learning, AI model training, data processing and AI applications.

Advantages, Limitations and Security

Advantages of Cloud Technology

- **Cost Efficiency** - Organizations can reduce the need for large investments in physical infrastructure.
- **Scalability** - Resources can be increased or decreased according to workload.
- **Accessibility** - Cloud services can be accessed from different locations through authorized devices.
- **Faster Deployment** - Cloud resources can be provisioned quickly.
- **Backup and Recovery** - Cloud services can simplify backup and disaster recovery.
- **Flexibility** - Organizations can select services according to their requirements.
- **Reliability** - Cloud providers can use redundant infrastructure to improve availability.

Limitations

- **Internet Dependency** — Cloud services often require a reliable Internet connection.
- **Security Risks** — Weak authentication or incorrect configurations can expose information.
- **Vendor Lock-in** — Moving applications between cloud providers can sometimes be difficult.
- **Downtime** — Cloud services can experience temporary outages.
- **Cost Management** — Uncontrolled resource usage can increase cloud expenses.

Cloud Security

Cloud security refers to the technologies, policies and practices used to protect cloud-based applications, infrastructure and data.

Important cloud security practices include:

- Strong passwords
- Multi-Factor Authentication
- Access control
- Encryption
- Regular updates
- Security monitoring
- Backup
- Least-privilege access

Shared Responsibility

Cloud security is generally a shared responsibility. The provider protects the underlying cloud infrastructure, while customers are responsible for protecting their accounts, applications, configurations and data according to the service being used.

Comparison, Future Trends and Conclusion

Comparison of Major Cloud Platforms

Feature	AWS	Microsoft Azure	Google Cloud
Provider	Amazon	Microsoft	Google
Compute	EC2	Virtual Machines	Compute Engine
Storage	S3	Blob Storage	Cloud Storage
Database	RDS	Azure SQL Database	Cloud SQL
Serverless	Lambda	Azure Functions	Cloud Functions / Run
Networking	VPC	Virtual Network	VPC
Strong Areas	Broad cloud services	Microsoft ecosystem	Data & AI

Future Trends

1. Edge Computing

Data is processed closer to the device or location where it is generated. This can reduce latency.

2. AI and Machine Learning

Cloud platforms are increasingly providing AI and ML services.

3. Serverless Computing

More applications are expected to use managed serverless services.

4. Multi-Cloud

Organizations may use multiple cloud providers according to their requirements.

5. Cloud-Native Development

Modern applications increasingly use containers, microservices, APIs, Kubernetes, automation and CI/CD.

6. Green Cloud Computing

Cloud providers are focusing on efficient resource utilization and energy-efficient infrastructure.